Area chart

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Use area charts to visualize monthly rainfall trends in 5 Indian cities

Step 1: Load the necessary libraries

```
library(ggplot2)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(readr)
library(tidyr)
library(stringr)
```

Step2:Load the rainfall data

```
rainfall_data <- read.csv("C:/Users/kumku/Downloads/dvfinalll.csv")
head(rainfall_data) ##Loads the rainfall data from a CSV file into R, so it can be analyzed
```

```
               STATE_UT_NAME DISTRICT   JAN  FEB  MAR   APR   MAY   JUN   JUL
1 ANDAMAN And NICOBAR ISLANDS  NICOBAR 107.3 57.9 65.2 117.0 358.5 295.5 285.0
2                   RAJASTHAN     TONK   5.0  5.3  2.7   3.0  10.1  56.1 219.7
3                   RAJASTHAN  UDAIPUR   2.8  2.0  4.3   2.2   8.2  73.6 207.5
4                   RAJASTHAN  DHOLPUR   8.6  9.5  7.0   2.5   8.6  54.5 211.7
5                   RAJASTHAN    BARAN   7.1  3.2  2.8   1.7   6.5  73.3 295.3
6                     GUJARAT   AMRELI   0.6  0.2  1.4   0.2   3.2 104.8 190.8
    AUG   SEP   OCT   NOV   DEC ANNUAL Jan.Feb Mar.May Jun.Sep Oct.Dec
1 271.9 354.8 326.0 315.2 250.9 2805.2   165.2   540.7  1207.2   892.1
2 210.5  79.7  13.9   4.9   3.0  613.9    10.3    15.8   566.0    21.8
3 203.7 106.5  15.2  11.1   2.6  639.7     4.8    14.7   591.3    28.9
4 266.1 117.7  26.3   4.3   5.3  722.1    18.1    18.1   650.0    35.9
5 303.6 120.0  25.2   9.7   4.3  852.7    10.3    11.0   792.2    39.2
6 126.2  96.4  25.4  16.6   1.5  567.3     0.8     4.8   518.2    43.5
```

Step3:Reshape data from Wide to Long format

Transform the data so that each row represents a single observation of rainfall for a specific district and month.

```
monthly_data <- rainfall_data %>%
  pivot_longer(cols = JAN:DEC, names_to = "Month", values_to = "Rainfall")
```

Step 4:Set the order of the months

Ensure that months are ordered chronologically for accurate plotting.

```
monthly_data$Month <- factor(monthly_data$Month,
                             levels = c("JAN", "FEB", "MAR", "APR", "MAY", "JUN","JUL", "AUG",
                             ordered = TRUE)
```

Step 5:Select specific cities for analysis

Choose the cities you want to analyze.

```
selected_districts <- c("Shimoga", "Udaipur", "Chennai", "Kolkata", "Thane")
```

Step 6:filter data for selected districts.

Subset the data to include only the selected districts

```
#Displays all unique district names in your monthly_data dataset.
unique(monthly_data$DISTRICT)
```

```
 [1] "NICOBAR"         "TONK"            "UDAIPUR"         "DHOLPUR"
 [5] "BARAN"           "AMRELI"          "BHAVNAGAR"       "JAMNAGAR"
 [9] "JUNAGADH"        "KUTCH"           "RAJKOT"          "SURENDRANAGAR"
[13] "PORBANDAR"       "DNH"             "DAMAN"           "DIU"
[17] "MUMBAI CITY"     "RAIGAD"          "RATNAGIRI"       "THANE"
[21] "SINDHUDURG"      "HYDERABAD"       "KARIMNAGAR"      "KHAMMAM"
[25] "MAHABUBNAGAR"    "MEDAK"           "NALGONDA"        "NIZAMABAD"
[29] "WARANGAL"        "RANGAREDDY"      "ANANTAPUR"       "CHITTOOR"
[33] "KUDDAPAH"        "KURNOOL"         "VELLORE"         "COIMBATORE"
[37] "DHARMAPURI"      "KANYAKUMARI"     "CHENNAI"         "MADURAI"
[41] "DHARWAD"         "GULBARGA"        "YADGIR"          "RAICHUR"
[45] "BAGALKOTE"       "GADAG"           "HAVERI"          "KOPPAL"
[49] "BANGALORE RUR"   "BELLARY"         "CHIKMAGALUR"     "CHITRADURGA"
[53] "KODAGU"          "HASSAN"          "KOLAR"           "MANDYA"
[57] "MYSORE"          "SHIMOGA"         "TUMKUR"          "BANGALORE URB"
[61] "CHAMARAJANAGA"   "DAVANGERE"       "RAMNAGAR(BNGR)"  "CHICKBALLAPUR"
[65] "ALAPPUZHA"       "CANNUR"          "ERNAKULAM"       "KOTTAYAM"
[69] "KOZHIKODE"       "MALAPPURAM"      "PALAKKAD"        "KOLLAM"
[73] "THRISSUR"        "THIRUVANANTHA"   "IDUKKI"          "KASARGOD"
[77] "PATHANAMTHITTA"  "WAYANAD"         "LAKSHADWEEP"
```

```r
# Convert both sides to lowercase to avoid mismatch
monthly_data <- monthly_data %>%
  mutate(DISTRICT = str_to_lower(DISTRICT))

selected_districts <- c("shimoga", "udaipur", "chennai", "tumkur", "thane")

rainfall_filtered <- monthly_data %>%
  filter(DISTRICT %in% selected_districts)# #Filters data to include only the 5 selected citie
#Removes any rows where rainfall is missing.

# View the result
#head(rainfall_filtered)

summary(rainfall_filtered$Rainfall)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.20    5.80   44.05  117.21  141.30  973.70
```

```r
rainfall_filtered %>% filter(is.na(Rainfall))
```

```
# A tibble: 0 × 9
# ℹ 9 variables: STATE_UT_NAME <chr>, DISTRICT <chr>, ANNUAL <dbl>,
#   Jan.Feb <dbl>, Mar.May <dbl>, Jun.Sep <dbl>, Oct.Dec <dbl>, Month <ord>,
#   Rainfall <dbl>
```

```r
rainfall_filtered <- rainfall_filtered %>%
  filter(!is.na(Rainfall))
```

Step 7:Plot the monthly rainfall trends

Create an area plot to visualize the monthly rainfall trends for each selected district

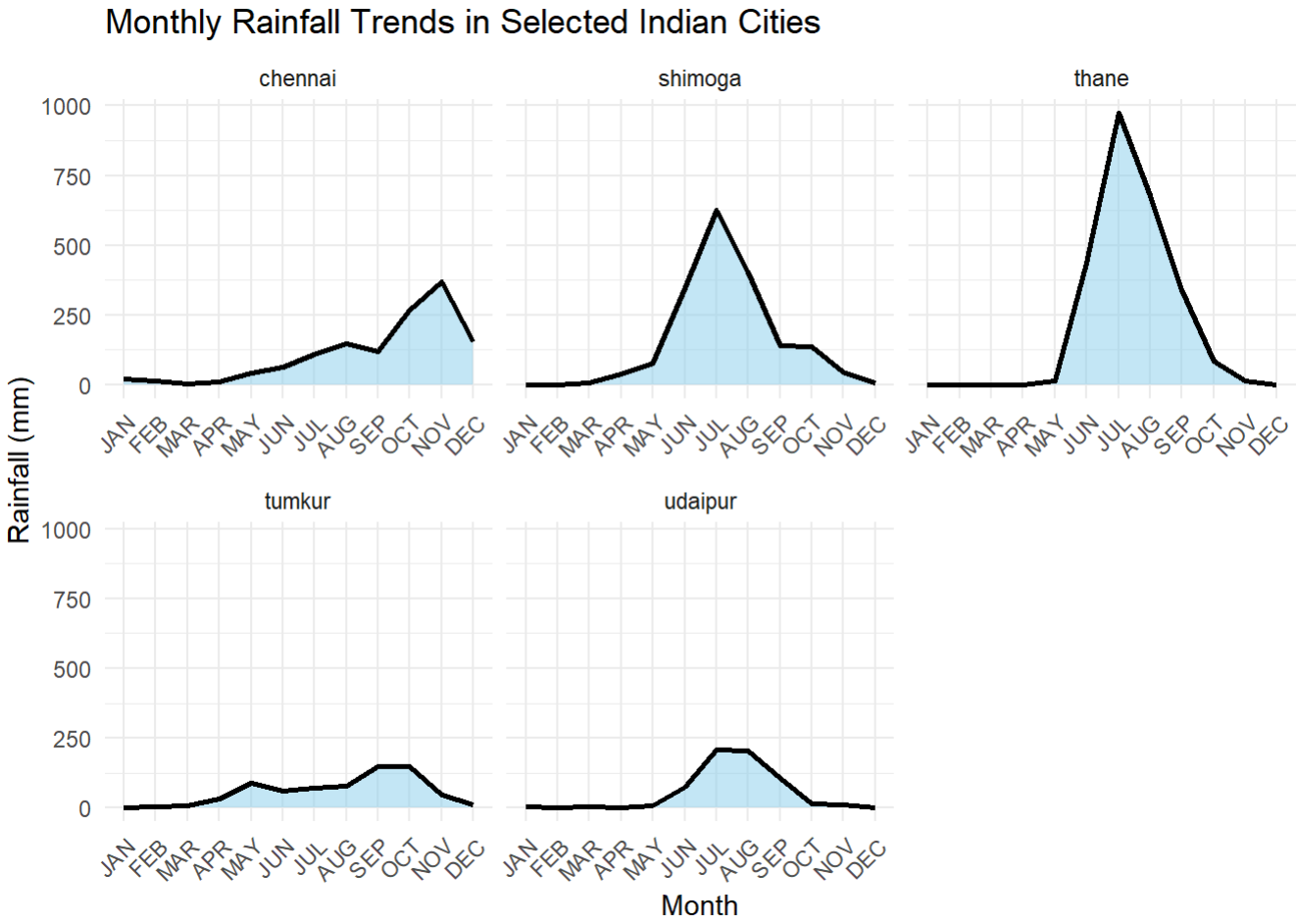
```r
# Ensure Month is a factor with all levels
rainfall_filtered$Month <- factor(
  rainfall_filtered$Month,
  levels = c("JAN", "FEB", "MAR", "APR", "MAY", "JUN",
             "JUL", "AUG", "SEP", "OCT", "NOV", "DEC"),
  ordered = TRUE
)

# Complete the data: fill in missing Month-DISTRICT combos
rainfall_completed <- rainfall_filtered %>%
  tidyr::complete(DISTRICT, Month, fill = list(Rainfall = NA)) # #If a month is missing for a
scale_x_discrete(drop = FALSE)
```

```
<ggproto object: Class ScaleDiscretePosition, ScaleDiscrete, Scale, gg>
    aesthetics: x xmin xmax xend
    axis_order: function
```

```
  break_info: function
  break_positions: function
  breaks: waiver
  call: call
  clone: function
  dimension: function
  drop: FALSE
  expand: waiver
  get_breaks: function
  get_breaks_minor: function
  get_labels: function
  get_limits: function
  get_transformation: function
  guide: waiver
  is_discrete: function
  is_empty: function
  labels: waiver
  limits: NULL
  make_sec_title: function
  make_title: function
  map: function
  map_df: function
  n.breaks.cache: NULL
  na.translate: TRUE
  na.value: NA
  name: waiver
  palette: function
  palette.cache: NULL
  position: bottom
  range: environment
  range_c: environment
  rescale: function
  reset: function
  train: function
  train_df: function
  transform: function
  transform_df: function
  super:  <ggproto object: Class ScaleDiscretePosition, ScaleDiscrete, Scale, gg>
```

```r
ggplot(rainfall_completed, aes(x = Month, group = DISTRICT)) +
  geom_ribbon(aes(ymin = 0, ymax = Rainfall), fill = "skyblue", alpha = 0.5) + #used to create
  geom_line(aes(y = Rainfall), color = "black", linewidth = 1) +
  facet_wrap(~ DISTRICT, scales = "free_x") +
  scale_x_discrete(drop = FALSE) + # #Prevents ggplot from dropping unused months from the x-a
  labs(
    title = "Monthly Rainfall Trends in Selected Indian Cities",
    x = "Month",
    y = "Rainfall (mm)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```


Monthly Rainfall Trends in Selected Indian Cities
chennai
shimoga
thane
tumkur
udaipur
Rainfall (mm)
1000
750
500
250
0
JAN
FEB
MAR
APR
MAY
JUN
JUL
AUG
SEP
OCT
NOV
DEC
Month